Tiger Sky Jewell-Alibhai

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Skills

Programming:
Micropython & Arduino C (platformio)
Python, Java
HTML & CSS

Tools/Manufacturing:
Advanced Wood Tools
3D Printers, Laser Cutter
Wire EDM
Mill, Shopbot CNC
Soldering
Composites techniques

Software:
SolidWorks, Onshape
Fusion 360
OrcaSlicer
SimScale FEA/CFD
Mission Planner/QGC
Ardupilot
Design Expert 13

Languages:
Mandarin, Portuguese

Other:
Licensed Drone Pilot (107)
AMA Member

tigrislabs.net

Hardware engineer, maker, and creator with experience designing, building, and controlling moving systems. Looking for positions in mechanical & robotics engineering.

Education

Olin College of Engineering Needham MA — ***Aug 2019 - May 2023***

- Bachelor of Science in Engineering, Concentration Robotics.
- Recipient of 4-year, 50% Franklin W. Olin College Merit Scholarship
- Courses in Mechanical Engineering, ECE, Programming, Lean-Agile Project Design, Applied Linear Algebra and Multivariable Calculus, Software Design, ODEs and Dynamic Systems, Mechanical Solids and Structures, and Mechanical and Aerospace Systems.
- Current cumulative GPA is 3.96

Durham Academy HS Durham NC — ***Aug 2015 - May 2019***

- Received Cole Award for Excellence in Physical Sciences (10th grade)
- Received Senior Science Award

Experiences

Autonodyne - Hardware R&D Engineer — ***September 2023 - Present***

- Design and production of companion computer systems for group 1 & 3 UAVs
- Design, fabrication, and testing of components for UAVs
- Flight testing of airframes and autonomous behaviours of UAVs

Ascent Aerosystems - Mechanical Engineer — ***May 2022 - Aug 2022***

- Designed and integrated an LTE control and video link into a prototype coaxial UAV.
- Design, integration, and documentation for manufacturing of UAV rotor assemblies and camera mounts.
- R&D flight testing and monitoring of coaxial UAVs.
- Maintenance of markforged and prusa 3D printers.

Olin Satellite Group - Mechanical Engineer — ***May 2021 - Aug 2021***

- Designed structural components in Solidworks for cross-university small satellite project involving the launch of three 3U satellites to collect atmospheric oxygen readings and demonstrate formation flying behaviour.
- Oversaw systems integration and prepared designs for manufacturing.

Practical Scientific Solutions - Mechanical Engineering Intern — ***May 2020 - Aug 2020***

- Iterated from prototype to final production design of ruggedized light emitting box.
- Designed a UAV protection system in Solidworks involving landing gear and prop guards.
- Lead and completed research projects into the use of radio and inertial tracking systems and the cost, technical, and logistical aspects of launching a cubesat.

Projects and Activities

Olin Design Build Fly Team - Project Manager — ***2019 - 2022***

- Project manager and former structures lead on Olin's AIAA Design Build Fly Project Team.
- Designing, testing, and flying of R/C planes for specific competitions.
- Experience with structural, electrical, and systems aspects.
- Lead the team to place 15th of over 100 teams at the AIAA DBF competition.

Olin Robotics Lab - Robotics & Software Engineer — ***2022 - Present***

- Developed and integrated indoor localization system using fiducial tags for ardupilot flight stack using mavros.
- Integrated and tested ultra-wideband indoor localization using the crazyflie control stack for a large scale indoor heavy lift quadcopter.
- Assisted and advised mechanical design, performed flight testing and PID tuning.